

C0859

Database Theory

Lecture 11

Normalization – Part 2

An Unnormalized Table

STU_NO	FEE	PAID	AMOUNT
302910	Co-op	Y	400.00
	Parking	N	150.00
	Site	Y	100.00
319825	Parking	Y	150.00
	Site	Y	100.00
327447	Site	Y	100.00
349223	Parking	Y	150.00
	Site	N	100.00

What would you do to put the table in 1NF?

Populate STU_NO

A 1NF Table

STU_NO	FEE	PAID	AMOUNT
302910	Co-op	Y	400.00
302910	Parking	N	150.00
302910	Site	Y	100.00
319825	Parking	Y	150.00
319825	Site	Y	100.00
327447	Site	Y	100.00
349223	Parking	Y	150.00
349223	Site	N	100.00

What would you do to put the table in 1NF?

Populate STU_NO

What is the key for the table when it is in 1NF?

STU_NO + FEE

Three Basic Requirements Of 1NF Table

- Cells must be single valued
- All values in a column must have the same domain
- Each row must be unique

STU_NO	FEE	PAID	AMOUNT
302910	Co-op	Y	400.00
302910	Parking	N	150.00
302910	Site	Y	100.00
319825	Parking	Y	150.00
319825	Site	Y	100.00
327447	Site	Y	100.00
349223	Parking	Y	150.00
349223	Site	N	100.00

1NF Anomalies

STU_NO	FEE	PAID	AMOUNT
302910	Co-op	Y	400.00
302910	Parking	N	150.00
302910	Site	Y	100.00
319825	Parking	Y	150.00
319825	Site	Y	100.00
327447	Site	Y	100.00
349223	Parking	Y	150.00
349223	Site	N	100.00

If the information on STU_NO 302910 is dropped, a deletion anomaly arises. What does this mean?

Information is lost -
Amount for Co-op

What other anomalies exist?

Insertion, Update

Second Normal Form

- A relation is in second normal form if all of its non-key attributes are dependent on all of the key

Normalize this 1NF table to 2NF.

What would it look like?

STU_NO + FEE -> PAID

FEE -> AMOUNT

STU_NO	FEE	PAID	AMOUNT
302910	Co-op	Y	400.00
302910	Parking	N	150.00
302910	Site	Y	100.00
319825	Parking	Y	150.00
319825	Site	Y	100.00
327447	Site	Y	100.00
349223	Parking	Y	150.00
349223	Site	N	100.00

Third Normal Form

- A relation is in 3NF if it is in 2NF and has no transitive dependencies

Verify that this table is in 1NF and 2NF.

What is the key for the table?

Where is the transitive dependency?

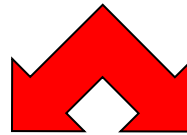
STU_NO	PROGRAM	TUITION
349207	Business	\$1,800.00
350194	Computer Systems Tech.	\$2,200.00
352896	Nursing	\$2,000.00
357104	Business	\$1,800.00

A deletion anomaly occurs when the second row is removed. What does this mean?

Normalize the table into two 3NF tables.

2NF to 3NF

STU_NO	PROGRAM	TUITION
349207	Business	\$1,800.00
350194	Computer Systems Tech.	\$2,200.00
352896	Nursing	\$2,000.00
357104	Business	\$1,800.00



PROGRAM	TUITION
Business	\$1,800.00
Computer Systems Tech.	\$2,200.00
Nursing	\$2,000.00

STU_NO	PROGRAM
349207	Business
350194	Computer Systems Tech.
352896	Nursing
357104	Business

Normalization Case Study Revisited

- Pipes N Things Inc.
 - A distributor of plumbing parts and fixtures
 - Purchase from manufacturers
 - Supply retailers
 - Designing corporate data base

Purchasing Department

- Determine vital entities
- Purchase Order document is critical
- This purchase order has a line number, which is simply a count of the number of items that appear on each particular purchase order
 - Some times this is known as a sequence number

List The Entities Found

Purchase Order

Date: 2024-07-15
P.O. #: 3067
Vendor ID: 507

Vendor: Crane Plumbing
572 Johnson St.
Willowdale, ON
M2M 3E4

Line	Qty	Item ID	Description	Unit Price	Line Total
1	80	P-796	Delta Cartridges	3.50	280.00
2	100	P-797	Moen Cartridges	3.25	325.00
3	25	P-993	Kitchen Faucets	18.75	468.75
Total					1,073.75

- Hint, look for keys
- Keys will usually identify entities

A Second PO

Purchase Order

Date:

2024-07-20

P.O. #:

3068

Vendor ID:

385

Vendor:

MCE Supply

12 Burris St.

Hamilton, ON

L8M 5R4

Line	Qty	Item ID	Description	Unit Price	Line Total
1	20	P-796	Delta Cartridges	3.50	70.00
2	50	P-853	2" O Rings	1.25	62.50
Total					132.50

Convert The Purchase Order Documents Into A Two Dimensional Table

PO_No	Date	Vend_ID	Name	Address	City	Prov	Postal	Line	Qty	Item_ID	Description	Price	Line_Total	Total
3067	2024-07-15	507	Crane Plumbing	572 Johnson St.	Willowdale	ON	M2M 3E4	1	80	P-796	Delta Cartridges	3.50	280.00	1073.75
								2	100	P-797	Moen Cartridges	3.25	325.00	
								3	25	P-993	Kitchen Faucets	18.75	468.75	
3068	2024-07-20	385	MCE Supply	12 Burris St.	Hamilton	ON	L8M 5R4	1	20	P-796	Delta Cartridges	3.50	70.00	132.50
								2	50	P-853	2" O Rings	1.25	62.50	

- Results in an un-normalized table
- Multiple occurrences of data exist
- We can find cells that have more than one value
 - Item_ID, Desc, Qty, Description, Price, Line_Total

Normalization – The First Step

- Transforming an un-normalized table into one that is in First Normal Form
- **1NF** has no repeated occurrences of data items in any of its cells
- Go back to the table and put it into 1NF by repeating some of the data values and removing calculated values

Normalization – The First Step

- Repeating Values For Attributes
 - PO_No, Date, Vend_ID, Name, Address, City, Prov, Postal
- Calculated values (Line_Total, Total) removed

PO_No	Date	Vend_ID	Name	Address	City	Prov	Postal	Line	Qty	Item_ID	Description	Price
3067	2024-07-15	507	Crane Plumbing	572 Johnson St.	Willowdale	ON	M2M 3E4	1	80	P-796	Delta Cartridges	3.50
3067	2024-07-15	507	Crane Plumbing	572 Johnson St.	Willowdale	ON	M2M 3E4	2	100	P-797	Moen Cartridges	3.25
3067	2024-07-15	507	Crane Plumbing	572 Johnson St.	Willowdale	ON	M2M 3E4	3	25	P-993	Kitchen Faucets	18.75
3068	2024-07-20	385	MCE Supply	12 Burris St.	Hamilton	ON	L8M 5R4	1	20	P-796	Delta Cartridges	3.50
3068	2024-07-20	385	MCE Supply	12 Burris St.	Hamilton	ON	L8M 5R4	2	50	P-853	2" O Rings	1.25

What is the table's key?

...PO_No + Line

Normalization – The Second Step

- Isolating
 - What data items are dependent on parts of the key
 - What data items are left dependent on the whole key

- Dependent on PO_No
 - Date
 - Vend_ID
 - Name
 - Address, City, Prov, Postal
- The resulting table
 - po_no -> date, vend_id, name, address, city, prov, postal

- Dependent on PO_No and Line
 - Qty
 - Item ID
 - Description
 - Price
- The resulting table
 - po_no + line -> qty, item_id, description, price

Normalization – The Third Step

- Separating any entity that deserves its own relation in the database
- Name each entity
- Look for a key field that has a dependent attribute; a key for another entity
- Is this occurring in our 2NF tables?
- If so, where?

Third Normal Form Tables

- po_headers
 - po_no -> date, vend_id
- vendors
 - vend_id -> name, address, city, postal, prov
- items
 - item_id -> description, price
- po_lines
 - po_no + line -> qty, item_id

Creating the ERD

- Draw the entities with their primary keys

po_headers	
PK	po_no
	date
	vend_id

vendors	
PK	vend_id
	name
	address
	city
	prov
	postal

po_lines	
PK	po_no
PK	line
	qty
	item_id

items	
PK	item_id
	description
	price

Creating the ERD

- Examine each primary key to see if it appears in another table, if so it is a foreign key in the other table

po_headers	
PK	po_no
	date
FK	vend_id

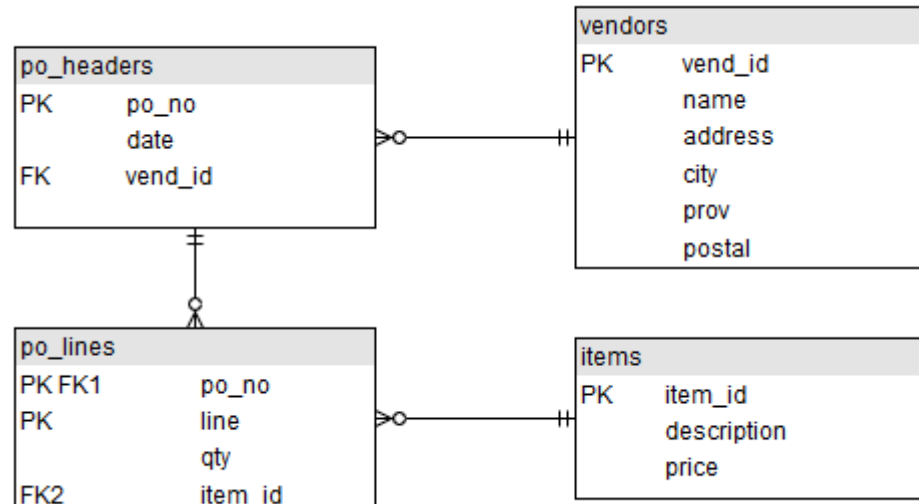
vendors	
PK	vend_id
	name
	address
	city
	prov
	postal

po_lines	
PK FK1	po_no
PK	line
	qty
FK2	item_id

items	
PK	item_id
	description
	price

Creating the ERD

- Each foreign key means there is a relationship between the two tables
- The primary key is on the “one” side and the foreign key is on the “many” side



Composite Keys

- There are some database designers that do not like composite keys
- An alternative is that any table that would have a composite key, get it's own unique key
- The column that would've been part of the composite key becomes just a foreign key
- This has emerged over the last years as the preferred design approach

Primary Key Types

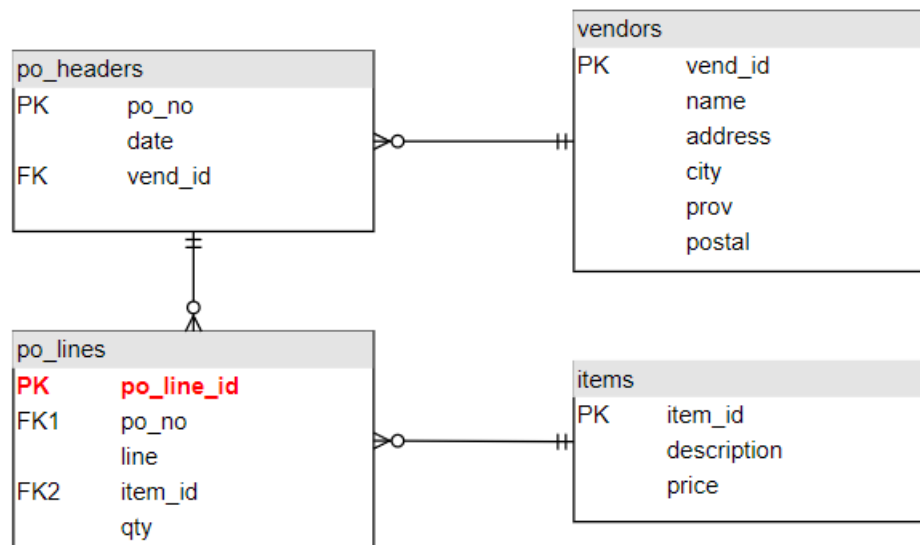
- There are two types of primary keys
 - Natural
 - Surrogate
- A natural key is one that “naturally” occurs in the data, like nursing unit id in CHDB
- A surrogate key does not naturally occur in the data and is usually a meaningless but unique number (MBUN)

Third Normal Form Tables - Alternative

- po_headers
 - po_no -> date, vend_id
- vendors
 - vend_id -> name, address, city, postal, prov
- items
 - item_id -> description, price
- po_lines
 - po_line_id -> po_no, line, qty, item_id

ERD

- The po_line table doesn't have a composite key anymore
- po_line_id is a surrogate key
- In fact, all of the primary keys in this ERD are surrogate



OLTP

- All examples considered thus far have been geared to Online Transaction Processing (OLTP)
- Most common
- Normalization rules promote unique (non redundant) data

Normalized Data

- Due to all of the joining required to return meaningful results, normalized data is complex to work with
- Derived columns cannot be indexed, therefore normalized data can be slow to work with
- For the sake of data integrity, in OLTP scenarios, these are tradeoffs we are willing to accept

OLAP and BI

- OLTP is not the only use for data
- Why would data be collected, if it wasn't going to be analyzed too?
- Online Analytical Processing (OLAP) is one of the other significant database uses
- Business Intelligence (BI) tools work with OLAP data to provide end users with the ability to analyze and report on the data in whatever way makes sense to the particular business

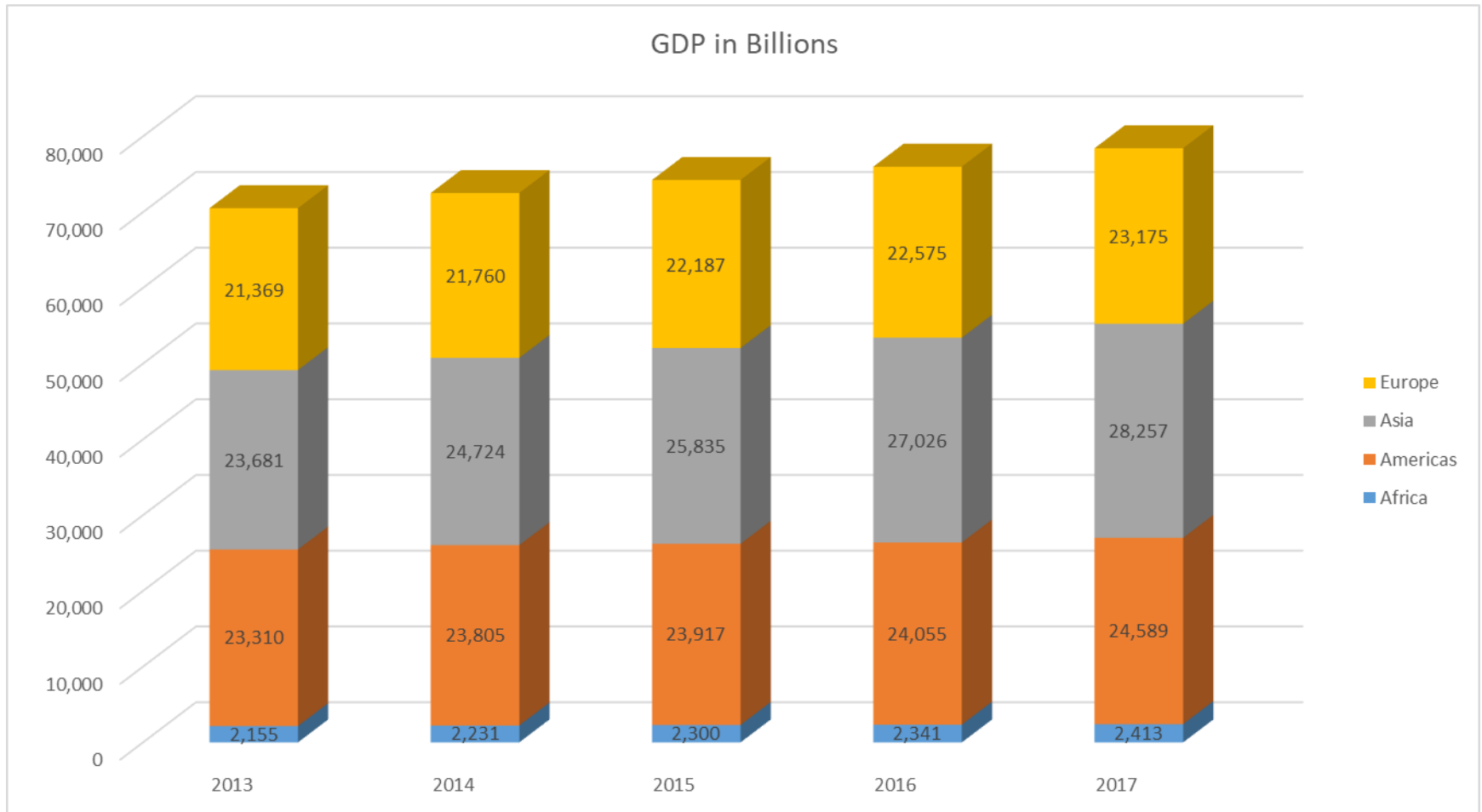
OLAP

- OLAP emphasizes speed and simplicity of access over data integrity
- In fact, OLAP databases are explicitly de-normalized for this purpose
- Data is also pre-summarized according to expected queries
- Also known as Big Data or a Data Warehouse

BI Tools

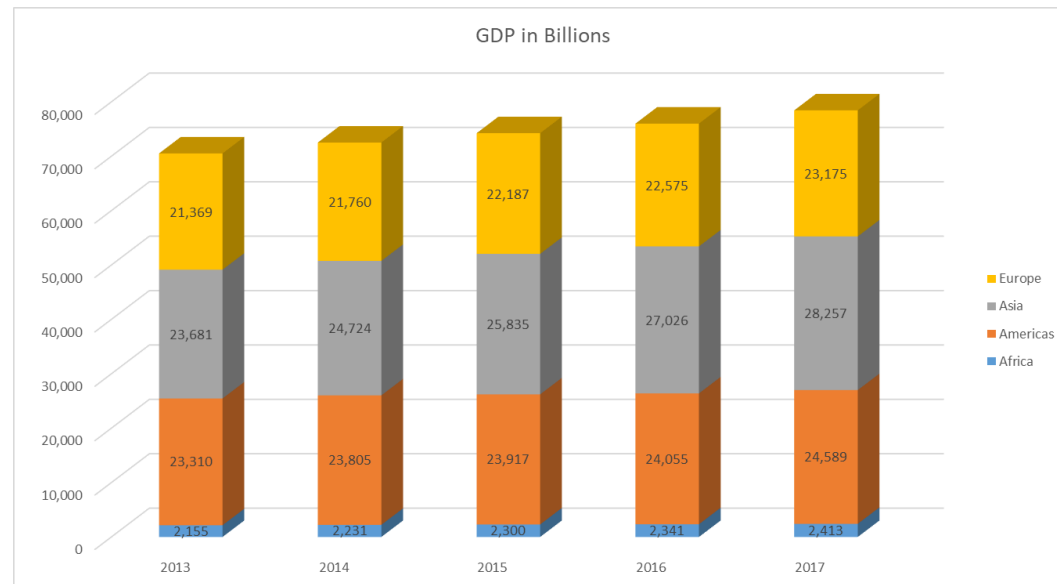
- Data in an OLAP database is often conceptualized around the concept of a cube, where one of the dimensions is time
- BI tools allow slicing, dicing, drilling down, etc.

Consider this Chart

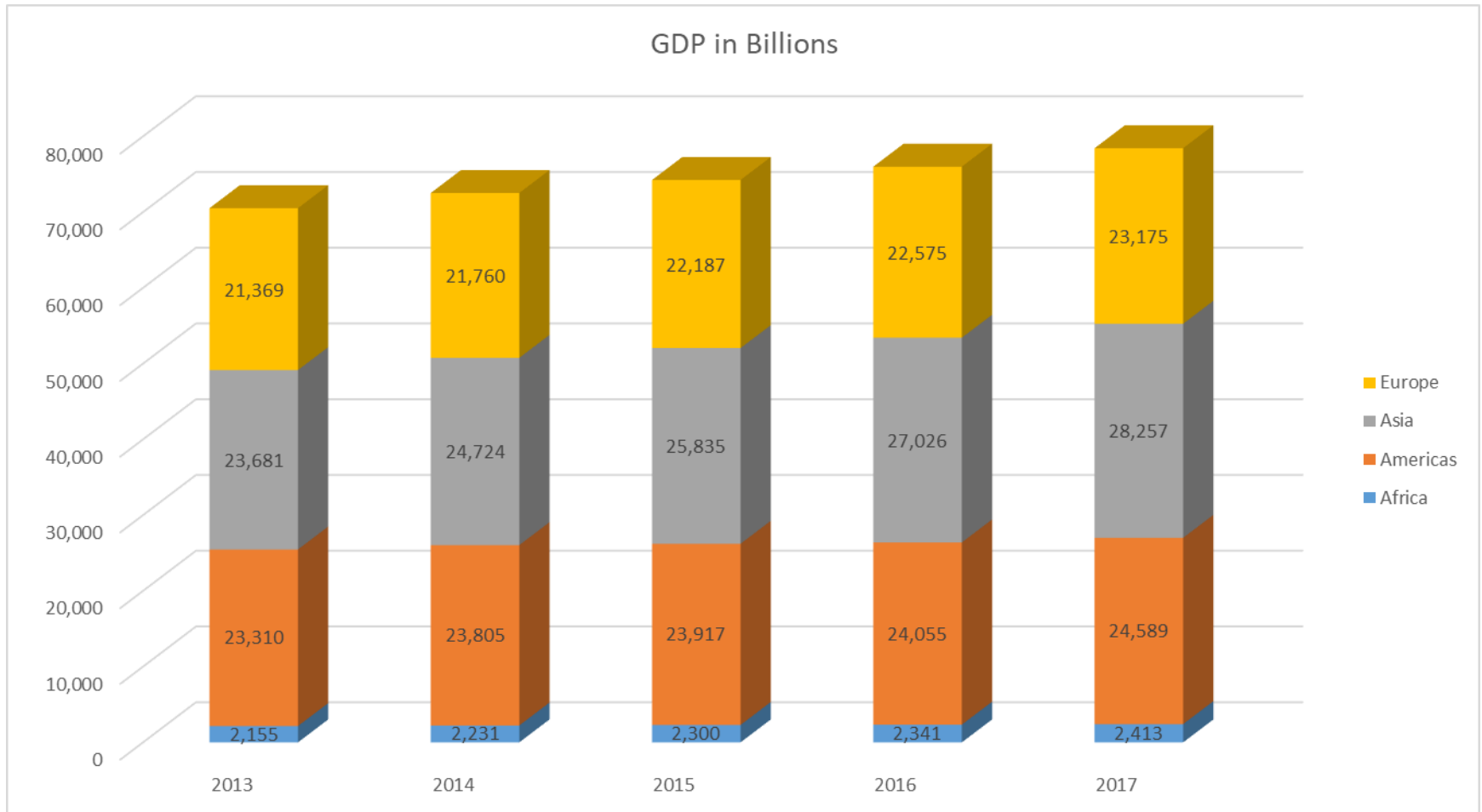


BI Simulation

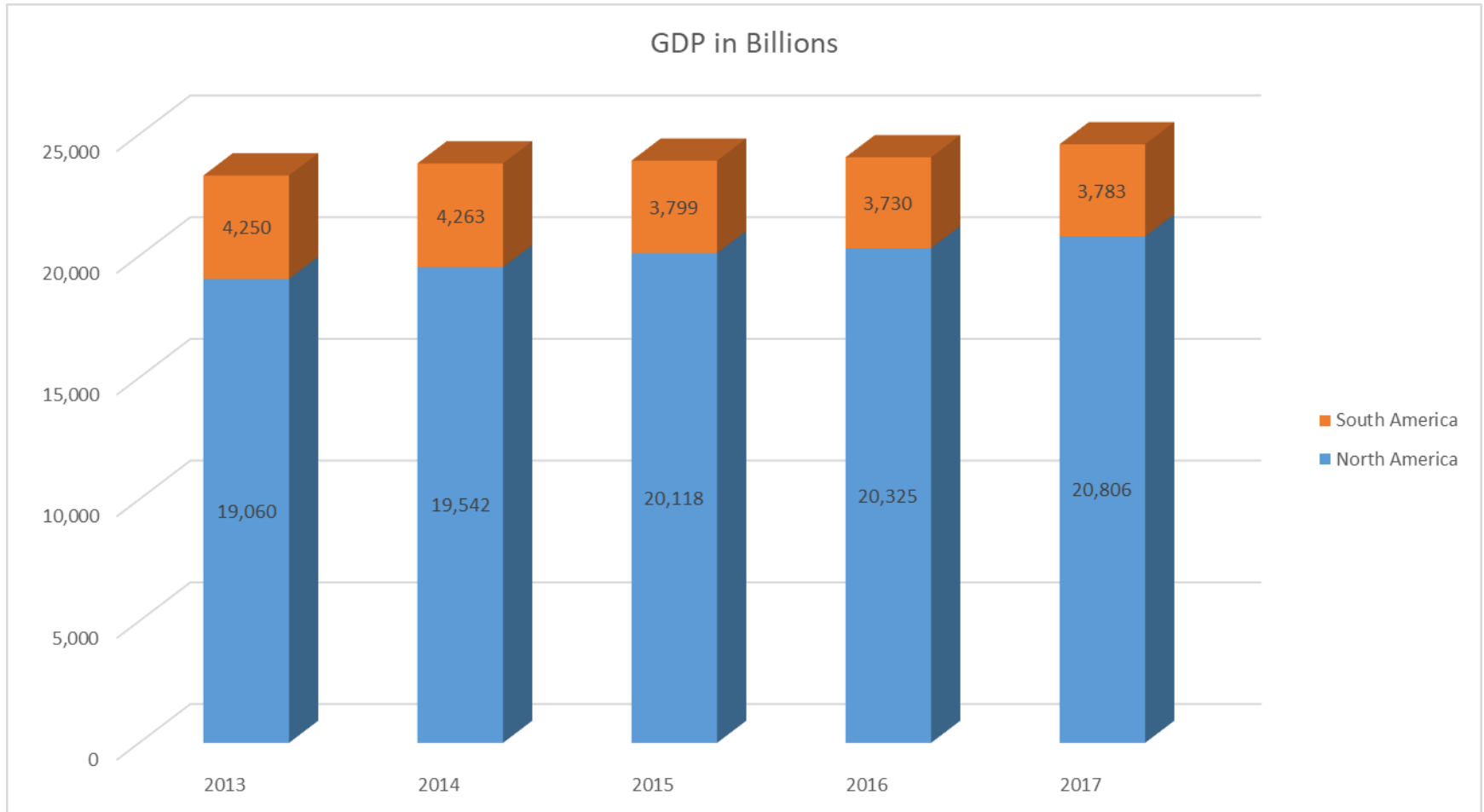
- In a real BI tool, you could click any element and “drill-down” to see more detail
- The following slides will present a simulation



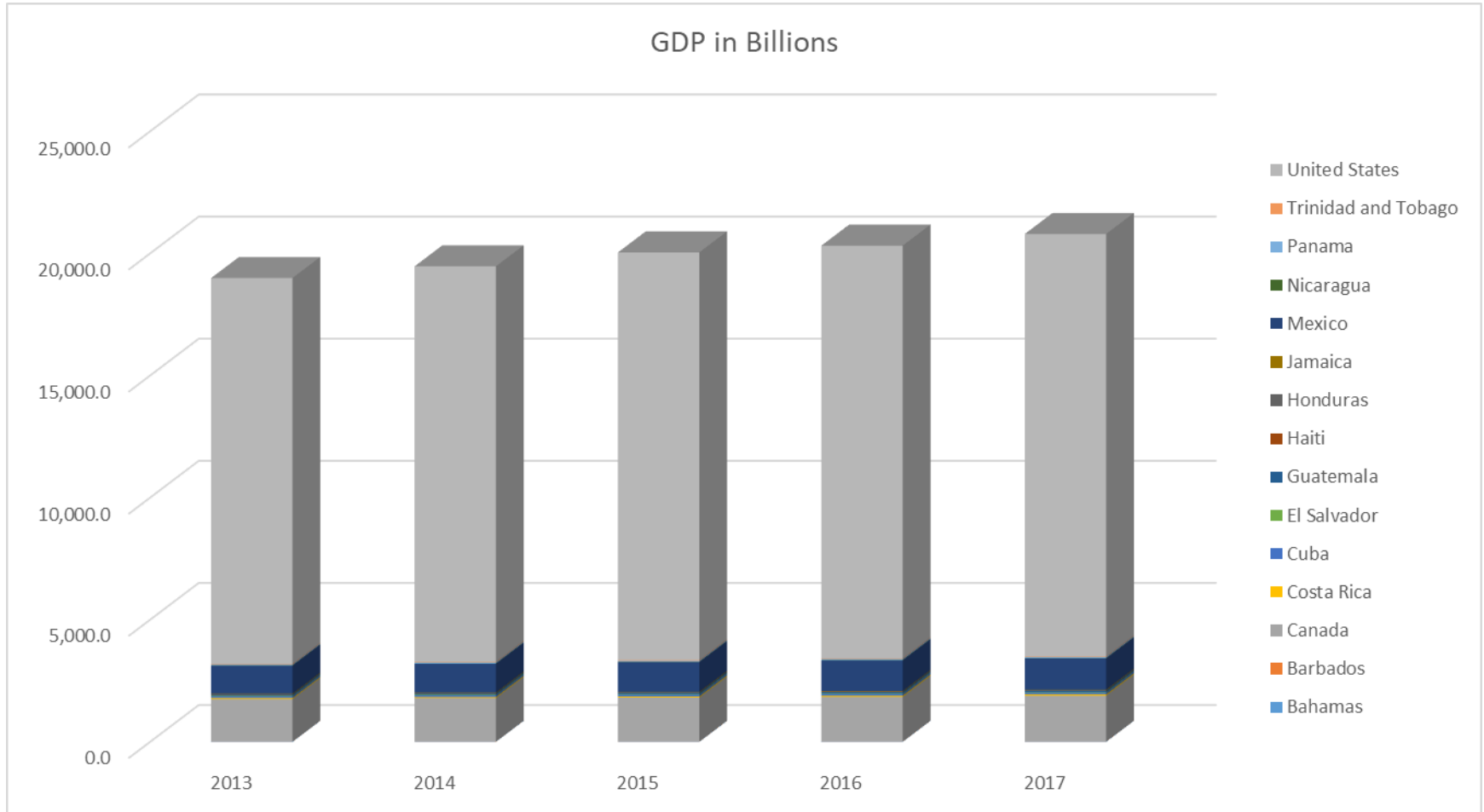
Click the Americas



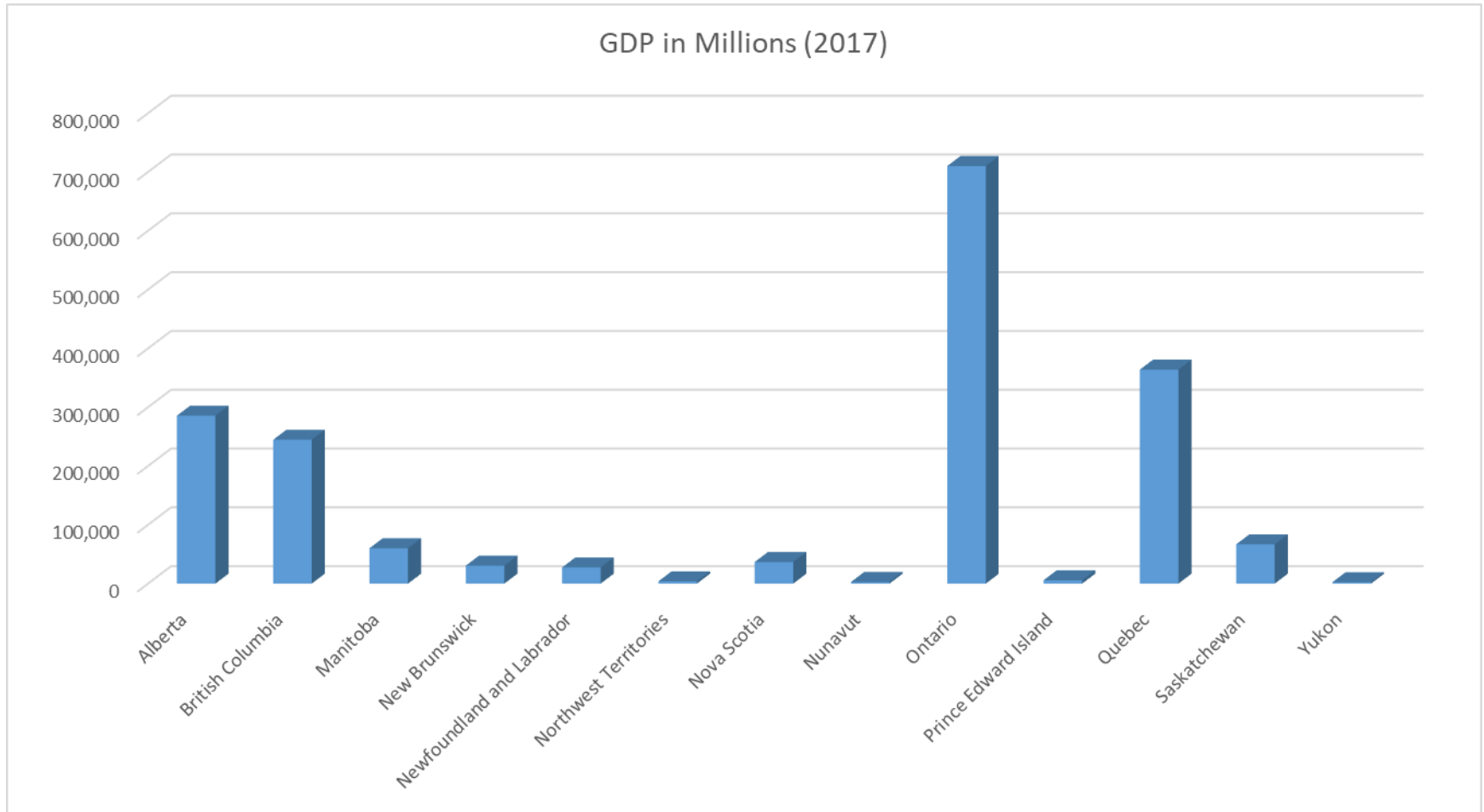
Click North America



Click Canada



Possible Final View of Data



Gapminder

- Gapminder is a Swedish non-profit venture
- It promotes sustainable global development
- It provides an excellent BI tool to analyze a number of statistics
- <https://www.gapminder.org/tools/>

Data Warehouse

- Offloading data from the OLTP to the OLAP database allows each database to be optimized for its particular task
- This separation ensures that complex, heavy inquiries into the OLAP database don't cause slow downs in the “production” database

Summary

- Normalizing data to 3NF is almost always adequate for “real world” processing
- OLAP de-normalizes and summarizes data for quick and easy analysis